MAT2002 **Name (Print):** _____
Spring 2026
HW1
Deadline: 1st Feb

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1. Determine the order of $\frac{d^2y}{dt^2} + \sin(t+y) = \sin t$ also state whether the equation is linear or nonlinear.

   Solution: Second Order; Nonlinear

2. Verify that $y = e^{t^2}\int_0^t e^{-s^2}ds + e^{t^2}$ is a solution of $y' - 2ty = 1$.

   Solution:
   $y' = 2te^{t^2}\int_0^t e^{-s^2}ds + 1 + 2te^{t^2}$
   $y' - 2ty = 2te^{t^2}\int_0^t e^{-s^2}ds + 1 + 2te^{t^2} - 2te^{t^2}\int_0^t e^{-s^2}ds - 2te^{t^2} = 1$

3. Solve the IVP: $y' - 2y = e^{2t}, \quad y(0) = 2.$

   Solution: The integrating factor is $\mu(t) = e^{-2t}$
   Then we obtain:
   $$e^{-2t}y(t) = t + c$$
   $$\big(\text{ or } y(t) = e^{2t}(t+c)\big)$$
   since $y(0) = 2$, we have
   $$y(t) = e^{2t}(t+2)$$

4. (Variation of Parameters) Consider the following method of solving the general linear equation of first order:
   $$y' + p(t)y = g(t).$$
   (a) If $g(t) = 0$ for all t, show that the solution is
   $$y = A\exp\left[-\int p(t)dt\right],$$
   where A is a constant.
   (b) If $g(t)$ is not everywhere zero, assume that the solution of Eq. (i) is of the form
   $$y = A(t)\exp\left[-\int p(t)dt\right] \tag{1}$$
   where A is now a function of t. By substituting for y in the given differential equation, show that $A(t)$ must satisfy the condition
   $$A'(t) = g(t)\exp\left[\int p(t)dt\right] \tag{2}$$

(c) Find $A(t)$ from Eq. (2). Then substitute for $A(t)$ in Eq. (1) and determine y. Verify that the solution obtained in this manner agrees with that of $y = \frac{1}{\mu(t)}\left[\int_{t_0}^{t} \mu(s)g(s)ds + c\right]$.

Solution:

(a)

$$\begin{aligned} y' &= -p(t)y \\ \Rightarrow \quad \frac{dy}{dt} &= -p(t)y \\ \Rightarrow \quad \ln|y| &= \int -p(t)dt + C \\ y &= A\exp\left[-\int p(t)dt\right] \end{aligned}$$

(b) $y = A(t)\exp\left[-\int p(t)dt\right]$
$y' + p(t)y = g(t)$
$\Rightarrow A'(t)\exp\left[-\int p(t)dt\right] - A(t)p(t)\exp\left[-\int p(t)dt\right] + p(t)A(t)\exp\left[-\int p(t)dt\right] = g(t)$
$\Rightarrow A'(t) = g(t)\exp\left[\int p(t)dt\right]$

(c)

$$\begin{aligned} A'(t) &= g(t)\exp\left[\int p(t)dt\right] \\ \Rightarrow A(t) &= \int g(t)\exp\left[\int p(t)dt\right]dt + C \\ y &= \exp\left[-\int p(t)dt\right]\left(\int g(t)\exp\left[\int p(t)dt\right]dt + C\right), \end{aligned}$$

which is exactly equivalent to Eq. (33).

5. Use the method above to solve the given differential equation $y' - 2y = t^2e^{2t}$.

Solution:

$$y' = 2y + t^2e^{2t}$$

Let $y = c(t)e^{2t}$, then

$$\begin{aligned} y' &= 2y + t^2e^{2t} \\ \Rightarrow & c'(t)e^{2t} + 2c(t)e^{2t} = 2c(t)e^{2t} + t^2e^{2t} \\ \Rightarrow & c'(t)e^{2t} = t^2e^{2t} \\ \Rightarrow \quad & c'(t) = t^2 \\ \Rightarrow \quad & c(t) = \frac{t^3}{3} + c \end{aligned}$$

So $\quad y = e^{2t}\left(\frac{t^3}{3} + c\right)$

6. Solve $\frac{dy}{dx} = \frac{x^2}{1+y^2}$.

Solution: The original equation can be written as:

$$\left(1 + y^2\right) dy = x^2 dx$$

Integrating both sides of the equation, we obtain:

$$y + \frac{y^3}{3} = \frac{x^3}{3} + c.$$

7. Solve the equation

$$\frac{dy}{dx} = \frac{ay + b}{cy + d},$$

where a, b, c, and d are constants.

Solution:

(1) If $a = b = 0$, then $y \equiv 0$.
(2) If $a = 0, b \neq 0$, then we have

$$\frac{cy + d}{b} dy = dx$$
$$\Rightarrow \frac{c}{2b} y^2 + \frac{d}{b} y = x + \tilde{c}$$

(3) If $a \neq 0$ and $ay + b \neq 0$, then we have

$$\frac{cy + d}{ay + b} dy = dx$$
$$\Rightarrow dx = \left(\frac{c}{a} + \frac{ad - bc}{a(ay + b)} \right) dy$$
$$\Rightarrow x + \tilde{c} = \frac{c}{a} y + \frac{ad - bc}{a^2} \ln |ay + b|$$

8. Suppose that a tank containing a certain liquid has an outlet near the bottom. Let $h(t)$ be the height of the liquid surface above the outlet at time t. Torricelli's principle states that the outflow velocity v at the outlet is equal to the velocity of a particle falling freely (with no drag) from the height h.

   (a) Show that $v = \sqrt{2gh}$, where g is the acceleration due to gravity.

   (b) By equating the rate of outflow to the rate of change of liquid in the tank, show that $h(t)$ satisfies the equation

$$A(h) \frac{dh}{dt} = -\alpha a \sqrt{2gh},$$

where $A(h)$ is the area of the cross section of the tank at height h and a is the area of the outlet. The constant α is a contraction coefficient that accounts for the observed fact that the cross section of the (smooth) outflow stream is smaller than a. The value of α for water is about 0.6 .

(c) Consider a water tank in the form of a right circular cylinder that is 3 m high above the outlet. The radius of the tank is 1 m , and the radius of the circular outlet is 0.1 m . If the tank is initially full of water, determine how long it takes to drain the tank down to the level of the outlet.

Solution: (a) since $\frac{1}{2}mv^2 = mgh$, then $v = \sqrt{2gh}$.
(b) Since the change of volumn of the liquid should be equal to the rate of outflow, we have

$$\begin{aligned} &\frac{dv}{dt} = -\alpha a v \\ \Rightarrow &\frac{dv}{dh}\frac{dh}{dt} = -\alpha a\sqrt{2gh} \\ \Rightarrow &A(h)\frac{dh}{dt} = -\alpha a\sqrt{2gh} \end{aligned}$$

(c)

$$\begin{aligned} &A(h)\frac{dh}{dt} = -\alpha a\sqrt{2gh} \\ \Rightarrow &\frac{1}{\sqrt{h}}dh = \frac{-\alpha a\sqrt{2g}}{A}dt \\ \Rightarrow &2\sqrt{h} = \frac{-\alpha a\sqrt{2g}}{A}t + \tilde{c} \end{aligned}$$

since when $t = 0, h(t) = 3$, we have

$$\tilde{c} = 2\sqrt{3}$$

Then $t = -\frac{(2\sqrt{h}-2\sqrt{3})A}{da\sqrt{2g}}$ When $h = 0$, we have

$$\begin{aligned} t &= -\frac{-2\sqrt{3}\cdot\pi}{0.6\pi(0.1)^2\sqrt{2\times 9.8}} \\ &\approx 130.4 \text{ s.} \end{aligned}$$

9. Suppose that a rocket is launched straight up from the surface of the earth with initial velocity $v_0 = \sqrt{2gR}$, where R is the radius of the earth. Neglect air resistance.

(a) Find an expression for the velocity v in terms of the distance x from the surface of the earth.

(b) Find the time required for the rocket to go $240,000$mi (the approximate distance from the earth to the moon). Assume that $R = 4000$mi.

Solution: Let x represent the height above the earth's surface. The equation of motion is given by

$$m\frac{dv}{dt} = -G\frac{Mm}{(R+x)^2},$$

where G is the universal gravitational constant. M and R represent the mass and radius of the earth, respectively. By chain rule,

$$\begin{aligned} mv\frac{dv}{dx} &= -G\frac{Mm}{(R+x)^2} \\ \Rightarrow \quad vdv &= -GM(R+x)^{-2}dx. \end{aligned}$$

Integrating both sides, and invoking $v(0) = \sqrt{2gR}$, the solution is:

$$v^2 = 2GM(R+x)^{-1} + 2gR - \frac{2GM}{R}.$$

since $g = GM/R^2$, we have

$$v(x) = \sqrt{2g}(R/\sqrt{R+x}).$$

(b) We now consider

$$\begin{aligned} \frac{dx}{dt} &= \sqrt{2g}\frac{R}{\sqrt{R+x}} \\ \Rightarrow &\sqrt{R+x}dx = \sqrt{2g}Rdt \end{aligned}$$

By definition of variable x, the initial condition is $x(0) = 0$. Integrating both sides, we have

$$x(t) = \left[\frac{3}{2}\left(\sqrt{2g}Rt + \frac{2}{3}R^{\frac{3}{2}}\right)\right]^{\frac{2}{3}} - R.$$

Setting the distance $x(T) + R = 240000$, and solving for T, we have $T \approx 50.4$ hours.

10. Solve $t^2y' + 2ty - y^3 = 0, \quad t > 0$

Solution: $t^2y' + 2ty = y^3$.

$$y + \frac{2}{t}y = \frac{1}{t^2}y^3.$$

multiply both sides by y^{-3} :

$$y^{-3}y' + \frac{2}{t}y^{-2} = \frac{1}{t^2}.$$
$$-2y^{-3}y' + \frac{-4}{t}y^{-2} = \frac{-2}{t^2}.$$
$$\frac{d}{dt}\left(y^{-2}\right) + \frac{-4}{t}y^{-2} = \frac{-2}{t^2}.$$
let $v(t) = y^{-2}(t)$.
$$\frac{dv}{dt} + P(t)v = Q(t).\text{where} \quad P(t) = \frac{-4}{t}, Q(t) = \frac{-2}{t^2}.$$
$$\mu(t) = e^{\int P(t)dt} = e^{-4\ln|t|} = t^{-4}$$
$$t^{-4}\frac{dv}{dt} + \frac{-4}{t^5}v = \frac{-2}{t^6}.$$
$$\left(\frac{v}{t^4}\right)' = \frac{-2}{t^6}.$$
$$\frac{v}{t^4} = \frac{2}{5}t^{-5} + c.$$
$$v = \frac{2}{5t} + ct^4$$
$$\frac{1}{y^2(t)} = \frac{2}{5t} + ct^4$$

$$y(t) = \pm\sqrt{\frac{1}{\frac{2}{5t} + c \cdot t^4}} = \pm\sqrt{\frac{5t}{2 + 5ct^5}}$$

11. Solve $y' = ry - ky^2, r > 0$ and $k > 0$.

Solution:

1. $y = 0$ is a solution.
2. For $y \neq 0$ Let $v = y^{-1}$ and $\frac{dv}{dt} = -y^2\frac{dv}{dt}$. So the differential equation becomes

$$-y^2\frac{dv}{dt} - ry = -ky^2$$

or $\quad \frac{dv}{dt} + rv = k.$

So $\mu(t) = e^{rt}$, and we have

$$v = \frac{k}{r} + ce^{-rt}.$$

since $v = y^{-1}$, we have

$$y = \frac{r}{k + cre^{-rt}}$$

12. Determine whether each of the following equations is exact. If it is exact, find the solution

(a) $(x \ln y + xy) + (y \ln x + xy)y' = 0; \quad x > 0, \quad y > 0$

(b) $\frac{x}{(x^2+y^2)^{3/2}} + \frac{y}{(x^2+y^2)^{3/2}} \frac{dy}{dx} = 0$

Solution:

(a)

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0.$$
$$M(x,y) = x \ln y + xy.$$
$$M_y = \frac{x}{y} + x$$
$$N(x,y) = y \ln x + xy$$
$$N_x = \frac{y}{x} + y$$

since $M_y \neq N_x$, it is not exact.

(b)

$$M_y = 2y \cdot \left(-\frac{3}{2}\right) \cdot \frac{x}{(x^2+y^2)^{\frac{5}{2}}}$$
$$N_x = 2x \cdot \left(-\frac{3}{2}\right) \cdot \frac{y}{(x^2+y^2)^{\frac{5}{2}}}.$$

since $M_y = N_x$. it is exact. By theorem. there exists a function $\psi(x,y)$ such that

$$\psi_x = M = \frac{x}{(x^2+y^2)^{\frac{2}{2}}}$$
$$\psi_y = N = \frac{y}{(x^2+y^2)^{\frac{3}{2}}}$$

Integrating ψ_x with respect to x gives

$$\psi(x,y) = \frac{-1}{(x^2+y^2)^{\frac{1}{2}}} + h(y)$$
$$\frac{\partial \psi(x,y)}{\partial y} = \frac{y}{(x^2+y^2)^{\frac{3}{2}}} + h'(y)$$
$$h'(y) = 0$$
$$\psi(x,y) = \frac{-1}{(x^2+y^2)^{\frac{1}{2}}}$$

Hence

$$\frac{-1}{(x^2+y^2)^{\frac{1}{2}}} = c$$
$$x^2 + y^2 = C^2 (C \in R).$$

13. Find the value of b for which the given equation is exact, and then solve it using that value of b.

(a) $\left(xy^2 + bx^2y\right) + (4x + y)x^2\frac{dy}{dx} = 0$

(b) $\left(ye^{2xy} + 5x\right) + bxe^{2xy}\frac{dy}{dx} = 0$

Solution: (a)

Let

$$M = \left(xy^2 + bx^2y\right)$$
$$N = (4x + y)x^2$$

When $b = 12$, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 2xy + 12x^2$, this equation is exact. There exists a function ψ, such that $\frac{\partial \psi}{\partial x} = M$ and $\frac{\partial \psi}{\partial y} = N$.

$$\psi = \int Mdx + h(y) = x^2y^2/2 + 4x^3y + h(y)$$

Since $\frac{\partial \psi}{\partial y} = x^2y + 4x^3 + h'(y)$, we have:

$$h(y) = \int 0dy = 0$$

The solution is

$$x^2y^2/2 + 4x^3y = C$$

(b)

Let

$$M = ye^{2xy} + 5x$$
$$N = bxe^{2xy}$$

When $b = 1$, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = e^{2xy}$, this equation is exact. There exists a function ψ, such that $\frac{\partial \psi}{\partial x} = M$ and $\frac{\partial \psi}{\partial y} = N$.

$$\psi = \int Mdx + h(y) = e^{2xy}/2 + 5x^2/2 + h(y)$$

Since $\frac{\partial \psi}{\partial y} = xe^{2xy} + h'(y)$, we have:

$$h(y) = \int 0dy = 0$$

The solution is

$$e^{2xy}/2 + 5x^2/2 = C$$